

SyncArms: Gaze-Driven Target Object-oriented Manipulation for Parallel Operation of Robot Arms in Distributed Physical Environments

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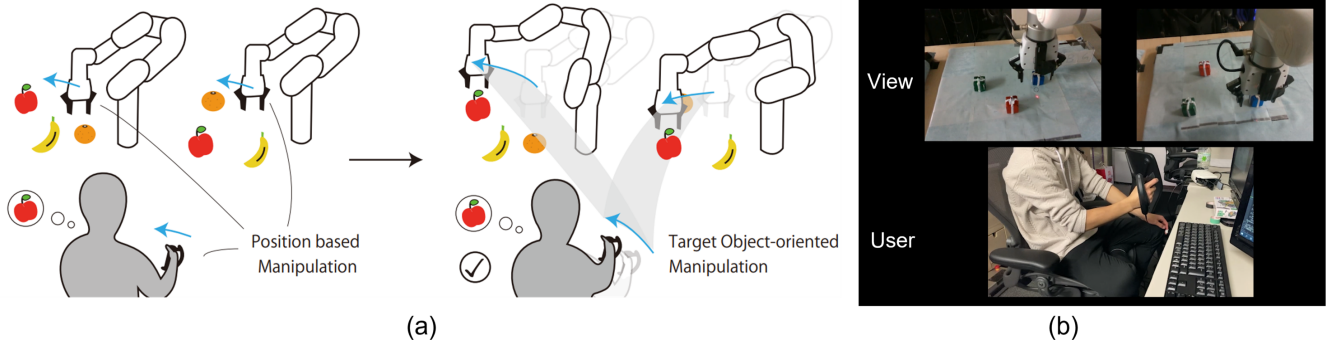


Figure 1: (a) The concept of our system for a parallel operation of synchronized multiple bodies. (b) User's view displayed on the monitor (upper two robot arm images), and a user manipulating the robot arm with the controller (lower image).

ABSTRACT

Enhancing human capabilities through the use of multiple bodies has been a significant research agenda. When multiple bodies are synchronously operated in different environments, the differences in environment placement make it difficult to interact with objects simultaneously. In contrast, if automatic control is performed to complement the differences and to perform a parallel task, the mismatch between the user and robotic arm movements generates visuomotor incongruence, leading to a decline in embodiment across the body. This can lead to difficulty completing tasks or achieving goals, and may even cause frustration or anxiety. To address this issue, we have developed a system that allows a parallel operation of synchronized multiple robotic arms by assisting the arm towards which the user's gaze is not directed while maintaining the sense of embodiment over the robotic arms.

CCS CONCEPTS

• Computer systems organization → Robotic control; • Human-centered computing → User interface design.

KEYWORDS

multiple bodies, gaze, parallel manipulation, robotic arm

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1 INTRODUCTION

Several studies have shown the possibility to enable multi-tasking by a single person. Direct mapping of the user's body information onto a robot or an avatar allows intuitive perception and manipulation. Synchronizing this behavior to multiple bodies makes it possible to perform the same or similar tasks in parallel in different locations[Miura et al. 2021]. Yet, if the environments are not aligned, the tasks cannot be completed in parallel due to differences in position. On the other hand, if an automatic control is performed to complement these differences in object placement, the automatic

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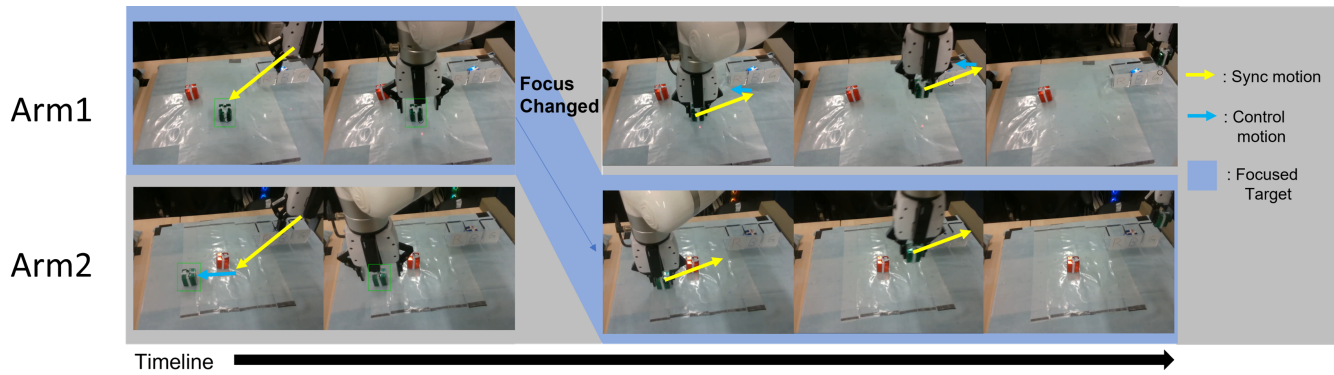


Figure 2: Example behavior when switching the focused target from Arm1 to Arm2. A focused target is always synchronized while a non-focused target is given complementary offsets.

motion causes an incongruence between the visually perceived motion of a robotic or virtual body and the actual motion that the user generated. As a consequence, the user’s embodiment is decayed due to the visuomotor incongruence [Takada et al. 2022]. To address this issue, we developed a system for synchronized operations of multiple robotic arms that implicitly provides positional assistance to the non-gazing arm. With this system, the user can perform the same task simultaneously in multiple environments without concern for the positional differences of the objects between environments.

2 ASSIST DESIGN

Our system was designed to minimize the user’s awareness of computer assistance. To account for differences between environments (with the gazing environment designated as the focused environment F , and the non-gazing environment as the non-focused environment NF), only the robotic arm in the environment NF was assisted by automatic control to allow for simultaneous operation in both environments. This ensured that the user always had visual-motor synchronization. Specifically, the assistance was provided by applying a spatial offset to the position of the robotic arm NF , such that the positions of each arm in the object coordinates of the target object being observed in environment F and the corresponding object in environment NF were the same for a horizontal x-y plane. The z-values are always synchronized. As the user may want to place the grabbed object in the same position with respect to the base coordinates, the offsets of each arm were made equal in each environment when the gripper of the robot arm was closed.

3 SYSTEM OVERVIEW

We used UFactory xarm as a robotic arm and Valve Index controller to map the user’s hand movements into the robotic arm with three DoF (translation). The system detected which environment and which object the user was looking at, by using the eye-tracking device Tobii pro nano. For the environments, as [Kasahara et al. 2016], the images of each environment were split and displayed on the monitor, with environment F determined by which side of the screen the user was looking at. We used YOLOv5 for object recognition. The target object was determined based on whether the user’s gaze point was within the detected bounding box for

400 ms, and its 3D position is computed by triangulation using the images from two cameras. In the demonstration, we will statically determine the object positions in advance for simplicity.

4 USER SCENARIOS AND DEMONSTRATION

Our gaze-based approach is particularly useful when you want to do a parallel task efficiently while keeping a sense of agency and interactivity. For example, with an implicit assistive system, parents raising multiple children can feed all of them simultaneously and feel that they fed them on their own. Besides, in a restaurant or a cafe, one may be able to plate several dishes in parallel and serve them to customers when receiving multiple orders at once.

In the demonstration, our system is constructed assuming the most basic of the actions described above: grabbing and placing. A user is asked to do a calibration for eye tracking. Then, the user holds a controller which is synchronized with robotic arm movement. With this controller, the user manipulates two robotic arms as Fig1(b) to perform pick-and-place tasks in parallel.

5 CONCLUSION

Our system allows users to manipulate multiple bodies in parallel, even in arbitrary situations and across multiple environments. Our gaze-driven approach is beneficial for designing parallel embodiment while maintaining the sense of agency and ownership for multiple bodies with machine intelligence.

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